



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

SQL PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Databases

TOTAL: 10 QUESTIONS

Q1 What type of database is SQL and what are its core strengths?

Threat Model

Q6 How do you monitor and tune slow queries in SQL?

Observability

Q2 How does SQL guarantee consistency and handle transactions?

Architecture

Q7 What backup and restore strategies does SQL support?

Lifecycle

Q3 What index types does SQL support and when do you use each?

Configuration

Q8 How do you handle schema migrations in SQL?

Compliance

Q4 How do you design a schema or data model in SQL?

Hardening

Q9 What security features does SQL provide?

Testing

Q5 How does SQL scale horizontally?

SSO / IdP

Q10 What are common anti-patterns when using SQL in production?

Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 SQL is a relational, document, key-value,

Mitigation

SQL is a relational, document, key-value, graph, or time-series store. Its strengths such as ACID guarantees, horizontal scale, or flexible schema guide the workloads it is best suited for.

A6 SQL provides a slow query log,

Observability

SQL provides a slow query log, explain/explain plan, or profiling tools to surface inefficient queries. Common fixes include adding indexes, rewriting queries, or increasing cache size.

A2 SQL provides either full ACID transactions

Primitives

SQL provides either full ACID transactions or eventual consistency depending on its CAP theorem positioning. Transaction support varies from none (simple K/V stores) to serializable isolation (relational DBs).

A7 SQL supports logical backups (export SQL

Automation

SQL supports logical backups (export SQL or BSON), physical backups (filesystem snapshot), and continuous replication to a standby. Point-in-time recovery requires WAL/oplog archiving on top of backups.

A3 SQL offers B-tree, hash, full-text, spatial,

Hardening

SQL offers B-tree, hash, full-text, spatial, or composite indexes depending on the engine. Choose based on query patterns: B-tree for range queries, hash for equality lookups, full-text for search.

A8 Schema migrations in SQL are managed

Governance

Schema migrations in SQL are managed with versioned migration files applied in order by a migration tool. Migrations should be idempotent and tested in staging before production to avoid downtime.

A4 Schema design in SQL involves normalizing

Gotchas

Schema design in SQL involves normalizing relations (relational DB) or denormalizing for access patterns (document/key-value). Understanding query needs upfront prevents costly migrations later.

A9 SQL supports role-based access control,

Assessment

SQL supports role-based access control, encrypted connections (TLS), encryption at rest, and audit logging. Always restrict user privileges to the minimum needed and disable anonymous access in production.

A5 SQL scales via replication (read replicas)

Federation

SQL scales via replication (read replicas) for read-heavy workloads and sharding (partitioning data by key) for write-heavy ones. Some SQL implementations handle this automatically; others require manual configuration.

A10 Common mistakes include selecting all

Optimization

Common mistakes include selecting all columns instead of needed fields, missing indexes on frequently queried columns, running migrations without backups, and storing large blobs directly in the database instead of object storage.